

Technical Memo: Server Rack Cooling Upgrade

To: Facilities Operations Team

From: Kyle Vance, Systems Administrator

Date: May 27, 2026

Subject: Installation of Smart Fan Trays in Server Rack 4B

I. Purpose

This memo outlines the performance results of installing an automated smart fan tray in Server Rack 4B to resolve ongoing equipment overheating issues.

2. Problem Status

Server Rack 4B houses high-density blade servers that regularly reached internal temperatures of 82°C, triggering automatic thermal throttling and slowing down network applications. The baseline ambient room cooling was insufficient for this specific rack.

3. Solution & Results

On May 12, 2026, a 1U smart fan tray equipped with automated thermal sensors was installed at the top of the rack. Temperature data was monitored over a 14-day period.

Metric	Pre-Installation	Post-Installation	Net Change
Average Rack Temp	82°C	68°C	14°C Reduction
Thermal Throttling Events	~5 per day	0 per day	100% Resolved
Fan Power Consumption	Constant (100%)	Variable (Avg. 45%)	55% Energy Saving

4. Conclusion

The smart fan tray successfully lowered internal rack temperatures into the safe operating zone (under 70°C) and completely eliminated performance throttling.

Next Step: Proceed with upgrading Racks 4C and 4D using the same hardware configuration during next week's maintenance window.